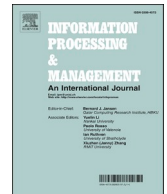




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Ordinal focal loss: A relative fashionability ranking learning method

Ling Ma^a, Chuhan Zou^b, Ziyi Guo^a, Tao Li^c, Zheng Liu^{a,d,e}, Fengyuan Zou^{a,c,d,*}^a Key Laboratory of Silk Culture Inheriting and Products Design Digital Technology, Ministry of Culture and Tourism, Zhejiang Sci-Tech University, Linping Campus, Hangzhou, Zhejiang 310018, China^b Amazon, 410 Terry Ave N, Seattle, WA, 98109, USA^c School of Fashion Design & Engineering, Zhejiang Sci-Tech University, Linping Campus, Hangzhou, Zhejiang 310018, China^d Clothing Engineering Research Center of Zhejiang Province, Zhejiang Sci-Tech University, Linping Campus, Hangzhou, Zhejiang 310018, China^e Zhejiang International Institute of Fashion Technology, Zhejiang Sci-Tech University, Linping Campus, Hangzhou, Zhejiang 310018, China

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ABSTRACT

Existing fashion recommendations often rely on natural language processing or content-based image retrieval, overlooking direct aesthetic assessments of fashion images. Given the subjectivity and complexity of this task, we propose treating fashionability as a relative attribute to rank paired clothing images. To address this ranking challenge, we propose Ordinal Focal Loss, which transforms the pairwise ranking problem into a multi-classification task, leveraging ordinal attributes to improve classification boundaries. Furthermore, in terms of fashion feature representation, we propose modeling not just individual items but also their combined effect as an outfit, providing a more holistic and nuanced fashion representation. We introduce the Fashionability3k dataset, comprising 3k image pairs (2398 ordered and 601 similar pairs) with objective relative fashion labels. Experiments on three datasets—our Fashionability3k and two public datasets—show that our method outperforms the baseline by nearly 1 % in ranking accuracy. In the user study, it achieved a 0.72 consistency with human subjective perception. Moreover, combining local and global visual features leads to additional performance gains, with an average improvement of 2.78 % in ordered pairs and 1.09 % in similar pairs. This is the first study to treat fashionability as an objective attribute for comparative analysis, validated through extensive experiments.

1. Introduction

Clothing is a basic necessity of life and an essential means of expressing culture and social interactions. However, matching clothes is not simple, especially today when fashion products are more abundant than ever. Surveys show that over 70 % of consumers struggle to make choices and lack fashion awareness (Hong et al., 2024). Fashion communities based on online shopping platforms, such as shoplook.io, fashiers.com, and wear.jp, are increasingly trying to improve this phenomenon. These fashion communities' primary feature is allowing users to share outfit-matching schemes and tips, whether or not they are fashion experts. As a result, the most

* Corresponding author at: School of Fashion Design & Engineering, Zhejiang Sci-Tech University, Linping Campus, Hangzhou, Zhejiang 310018, China.

E-mail addresses: 202110201019@mails.zstu.edu.cn (L. Ma), zouchuha@amazon.com (C. Zou), 202210201006@mails.zstu.edu.cn (Z. Guo), lt007@zstu.edu.cn (T. Li), koala@zstu.edu.cn (Z. Liu), zfy166@zstu.edu.cn (F. Zou).

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significant benefit of the cooperation between fashion communities and online shopping platforms is that many users will choose to buy complete outfits because they believe it saves time and is fashionable. The objective of this study is to assess the relative fashionability of outfits based on user aesthetic preferences to facilitate complete outfit transactions.

Benefiting from the fashion community, we have many images of items or outfit sets that can serve as research subjects in the fashion domain. Most of the research, which primarily focuses on **content-based retrieval** (Baldrati et al., 2022; Lu et al., 2023), **compatibility modeling** (Balim & Özkan, 2023; Mo et al., 2022; Moosaei et al., 2022; Sarkar et al., 2023; Wang et al., 2022) and **aesthetic assessment** (Alshan et al., 2018; Yu et al., 2018; Zou et al., 2022), is motivated by future applications in recommendation systems aimed at enabling more accurate recommendation of items and outfits. Before conducting further research, it is necessary to analyze how this proposed task differs from content-based retrieval and compatibility modeling, thereby underscoring the necessity of this study.

Content-based fashion retrieval has evolved from the initial similar-product retrieval to today's matching-item retrieval. However, it still lacks the ability to directly relate to users' subjective aesthetic preferences—that is, the capability to rank the fashionability of outfits. The main reasons are: (1) Different Objectives: Although both content-based retrieval and fashionability ranking involve image embeddings, they address entirely different problems. Content-based retrieval typically aims to analyze the visual features of new product images to identify similarities with the query content (Islam et al., 2024). In contrast, fashionability ranking focuses on evaluating the aesthetic value of outfit combinations. It is not a simple similarity comparison; instead, it aims to estimate an appropriate quantitative score that aligns with human perception (Chounchenani et al., 2025). (2) Complex Ground Truth: Fashionability ranking corresponds to a more complex ground truth. The ground truth for content-based retrieval is usually defined by objectively similar products. In comparison, fashionability assessment involves reflecting human subjective aesthetics, requiring the model to establish effective connections between diverse features and subjective visual impressions. For example, Suvama and Balakrishna (2024) points out most previous studies rely on recommendation loss alone to guide visual understanding and matching; that is, the features captured by these methods describe basic characteristics (e.g., color, texture, shape) of the input items, they are not directly related to the visual signals of the output items (to be recommended). Consequently, we further summarize reason (3) as follows: Compared with fashionability ranking, content-based retrieval faces limitations in objective function design. Traditional content-based retrieval methods typically depend on distance metrics (e.g., Euclidean distance) or contrastive loss (Islam et al., 2024), which are essentially simple mapping functions. In contrast, fashionability ranking requires both evaluation and comparison, which means that its objective function must consider the ambiguous correspondence between embedding features and the ground truth as well as complex nonlinear interactions. For example, to visually support the above analysis, we compare this study with OutfitTransformer (Sarkar et al., 2023) as shown in Table 1.

Compatibility modeling represents how compatible when two or multiple items are together. If two items are frequently combined together in high-quality outfits, then they can be deemed as compatible, otherwise not (Han et al., 2017; Song et al., 2017; Vasileva et al., 2018). However, higher compatibility does not necessarily imply higher fashionability. The challenge for future research lies in extending these general methods—such as metric learning, sequence learning, autoencoders, and graph networks—to personalized applications that prioritize users' subjective aesthetics (Ding et al., 2023; Selwon & Szymański, 2024).

Fashion aesthetic assessment has potentially very positive implications for practical applications (Rodriguez-Fernandez et al., 2022) (generally, users' decision depends mainly on whether the clothing is in line with their aesthetics), aesthetic assessment based on fashion images should also be an essential component that recommendation systems need to consider. Due to its subjectivity, it isn't easy to directly establish a mapping relationship between it and images. People often waver over which of several outfit combinations is more beautiful, not to mention computers. But an interesting phenomenon arises here: Although humans can quickly make binary

Table 1

Comparison between OutfitTransformer and this study.

	The proposed task	OutfitTransformer
Subject	Outfits pair: 	Outfit items: 
Label	Relative fashionability 2: more; 1: equal; 0: less	Compatibility 1: compatible; 0: incompatible
Objective	Ranking fashionability of two outfits	Computing items compatibility score
Approach	Features of outfits image, items image and text→ outfit representation→ difference learning→ ordinal multiclassification	Features of items image and text→ global outfit representation→ outfit compatibility score→ binary classification
Function	$r = \text{softmax}(\text{OrdinalClassifier}(\Delta(x_{\text{outfit}_i}, x_{\text{outfit}_j})))$	$c = \text{MLP}(E_{\text{trans}}(X_{\text{outfit}}, F))$

choices based on subjective preferences, it becomes difficult to rank multiple options. Conversely, computers can capture general patterns through large-scale supervised learning but struggle to grasp the nuanced preferences behind binary choices.

Consumer behavior research indicates that people typically choose 2 to 3 products from a recommendation list to add to their shopping cart in order to make the final purchase decision (Wang et al., 2022). This aligns with the Cognitive Load Theory (Sweller, 2024). Thus, it is expected that if recommendation systems could assist users in making final purchase decisions more quickly based on image aesthetics, i.e., fashionability, it would further enhance the user experience (see Fig. 1 for an illustration). In other words, we are now faced with the problem of comparatively evaluating the fashionability of pairs of images.

Treating the fashionability comparison between image pairs as a binary classification problem is overly absolute and arbitrary, as considering fashionability as an absolute attribute with values of 0 or 1 would be excessively contrived. In contrast, viewing fashionability as a relative attribute would be more accurate, and this accuracy can stem from the data collection stage since determining if one outfit is more fashionable than another is more accessible than directly assessing the fashionability of a single outfit, aligning with human subjective cognition.

To this end, this study proposes an ordinal focal loss, derived from the softmax-based multi-class version of focal loss, which incorporates ordinal information to compare and rank outfit fashionability. The objective of optimizing the loss function is to maximize classification accuracy. Consequently, it is crucial to establish comparative relationships in fashionability among samples. However, existing public datasets primarily provide absolute assessments for each outfit. Although we applied a label conversion technique to transform the absolute fashionability labels in public datasets into relative ones for subsequent model generalization experiments, to better capture subtle relative differences in fashionability, we created a dataset named Fashionability3k. Additionally, at the outfit-level representation stage, our approach posits that using outfit-level representations (referred to as global representations in this study) is more effective in capturing the complex feature interactions among multiple items within an outfit, compared to simply concatenating the visual features of individual items (referred to as local features in this study). Therefore, we propose a joint local-global feature representation for outfits. Lastly, results from comparative experiments and human approval tests suggest that the strategy designed has valuable applications in the industry. Briefly, our contributions are:

- To our knowledge, the idea of using relative attribute learning to rank the fashion of clothing matching has not been explored in any prior work.
- Customized objective function that describes the logic of the proposed relative fashion ranking method based on the first contribution.
- Experimental evaluation and analysis of our proposed method that achieves competitive results.
- A new complete outfit dataset, along with relative fashion labels, is released to the community for image aesthetic assessment.

This paper is organized in the following manner. Section 2 discusses some of the related work. Section 3 describes the approach of this paper in tackling this problem. Section 4 introduces a new dataset (Fashionability3k), presents our experiments, and validates them through comparison. Section 5 outlines our work's theoretical and practical implications. Section 6 concludes the paper and discuss the limitations.

2. Related works

Relative attributes: Relative attributes (Parikh & Grauman, 2011; Wang, 2024) expand the domain of attribute values from a binary spectrum to a continuous scale ($-\infty, +\infty$). This transition addresses the inherent ambiguity and limitations of absolute attribute values, facilitating a more precise conveyance of semantic attribute information. The paper elucidates two promising avenues for applying relative attribute learning: zero-shot image classification and the generation of advanced semantic representations for image descriptions. In the domain of image description, the aim is to characterize specific attributes of novel images through the comparative intensity of those attributes discerned within a training dataset (Ahmed & Yanikoglu, 2023). Specifically, Shehata et al. (2023) highlight the importance of delineating essential gait attributes (stride length, frequency, and width) for enhancing gait recognition accuracy via an uncertainty-aware model that estimates relative gait attributes. Furthermore, leveraging the superior capability of relative attribute descriptions in image quality assessment tasks, this approach aims to reconcile computational and human subjective perceptions of abstract concepts like beauty. Altwaijry and Belongie (2013) explore facial attractiveness by implementing a learning model for relative facial attractiveness ranking, effectively identifying key features contributing to perceived beauty, thereby facilitating accurate attractiveness prediction for new faces. As research progresses, more studies highlight the relative



Fig. 1. A recommendation scenario incorporating outfit fashionability ranking.

importance of different visual features in aesthetic evaluation (Li et al., 2020; Yan et al., 2024). For example, Huang et al. (2024) suggests that the weight of attributes, from coarse to fine, better predicts an image's aesthetic quality. This aligns closely with our exploration of how global outfit features and individual garment details contribute separately and together to the perception of fashionability.

Kovashka (2012); Yu and Grauman (2014) leverage relative attributes for fine-grained description retrieval in shoe shopping, emphasizing attributes like sportiness and comfort by assessing detailed features (material, heel height, opening size) to infer subjective perceptions, providing insightful methodologies for fashion attribute analysis. However, these studies primarily focus on ordinal comparisons, neglecting attribute similarities, which may conflict with human subjective assessments when items are perceived as equally fashionable.

Fashion aesthetic assessment: Fashion aesthetics evaluation has evolved into a critical research domain within the apparel and fashion industry, focusing on the aesthetic appraisal of fashion items. With the integration of digital technologies, the field has transitioned from subjective judgments to objective, quantitative methodologies. Utilizing advancements in computer vision and artificial intelligence, researchers can automatically analyze fashion imagery elements—color (Adilova & Shamoi, 2024; De Divitiis et al., 2023), texture (Cui et al., 2021; He et al., 2024) and silhouette (Cui et al., 2021; De Divitiis et al., 2023)—to assess their collective impact on aesthetic outcomes. This shift enables more nuanced consumer preference studies and more innovative recommendation systems. They basically do the aesthetic feature extracted by the pre-training on the network, which is a brain-inspired deep structured trained (Wei Gong, 2021) for the assessment task of aesthetics. But then, Suvarna and Balakrishna (2024) points out most previous studies rely on recommendation loss alone to guide visual understanding and matching. Addressing this oversight is crucial, as establishing a direct connection between input and output is a fundamental prerequisite for training a model, particularly within the recommendation domain. We aim to bridge this gap by associating feature inputs with user aesthetic preferences, enhancing the model's relevance and predictive accuracy in aligning with individual tastes.

Linking it with users' aesthetic preferences is inevitable for recommendation systems to become more intelligent and personalized, as user's decisions are driven not only by textual features but also by the aesthetic features of products. Based on this, Sun et al. (2024), Zhou et al. (2021) developed composite neural network architecture that captures actual fashion products' aesthetic and descriptive features. Although, from their experiments, title-based features are insufficient to make sound recommendations, results have shown that combining image features can yield better performance than independently. This conclusion also supports our simultaneous collection of images and text descriptions as experimental data. In addition, incorporating user style into recommendations (Kachbal et al., 2024) has shown promising results in mitigating the bias towards popular items, highlighting the significance of aligning recommendations with individual aesthetic preferences. Similarly, Liu et al. (2021); Xu et al. (2024) demonstrated that aesthetics (customers' style preferences) can be inferred from their online shopping behaviors and the products they have expressed interest in.

However, the recommendation task based on users' fashion aesthetic preferences cannot be separated from subjective comments in the data collection phase, which largely affects the performance of machine learning models trained from annotated data. To reduce the rating uncertainty, Gonzalez-Naharro et al. (2023) solve the problem by re-weighting the most frequent and infrequent ratings. Lv et al. (2023) constructs a personalized image collection platform for image aesthetic enhancement and assessment. Volunteers are asked to score and retouch the images based on their preferences. Likewise, we built a simple platform during the data collection stage to automatically collect subjects' fashion comparison relationships between two sets of clothing images. Compared with arbitrary thresholds like 5 in 1–10 ratings, the advantage of relativity appears.

Ordinal classification: In fashion recommendation research, ordinal classification leverages machine learning and AI to rank fashion elements, enhancing personalization and aligning with user preferences. Unlike traditional classification, ordinal classification (Shah et al., 2024; Singer et al., 2020) acknowledges natural order among categories, essential for sorting attributes like outfit suitability or trendiness.

At the application level, in order to predict the purchasing in the fashion field (substantial, middle, and inconsiderable), Fallah Tehrani and Ahrens (2016) proposed a novel method based on kernel machine regression equipped with the ordinal class classification setting. In the specific implementation, the threshold models are used, that is, "cut points," as mentioned. Tej et al. (2023) also introduces the ordinal problem of fashion products corresponding to size recommendations. Accurate recommendation is an important prerequisite to ensure product sales. To further emphasize users' interest preferences, Lee et al. (2024) considered it more natural to view user ratings as ordinal. In this way, they can map from user-assigned discrete rating scores to the exact magnitude/degree of user preferences for items. A literature review reveals an apparent absence of studies treating fashionability levels as an ordinal classification problem. Given the subjective ambiguity and complexity of it, this oversight is understandable. However, fashionability levels can reflect the aesthetic value of clothing and outfits, offering potential significant value for personalized recommendation systems. Moreover, considering fashionability levels as an ordinal classification problem enables the use of more nuanced models to understand the relative differences among various fashion elements beyond merely categorizing them as "trendy" or "not trendy."

Addressing ordinal classification necessitates exploiting the ordinal relations among categories to boost model accuracy and robustness while being mindful of category finiteness and adjacent class distance uncertainty. This contrasts with typical multi-class scenarios (distinguishing between dresses, pants, etc.) lacking ordinality, prompting researchers to implicitly integrate ordinal constraints in the feature space. Diaz et al. (2024); Shah et al. (2024) put forward methods where target labels are recoded into vectors, representing the total number of categories. Each dimension k of the vector addresses a specific question: "Is the target ranking greater than $k - 1$ and less than $k + 1$?" This encoding strategy effectively defines an interval rather than a single point for each category, aiding the model in better understanding and learning the relative positional relationships within ordinal data. Nousi and Tefas (2024), Yu et al. (2024) softened the one-hot label vectors into a label distribution with a unimodal shape, reflecting the natural ordinal relationships between categories. A unimodal distribution implies that the label distribution is concentrated around a central point and

decreases towards both sides. Additionally, the issue of ordinal classification can be converted into $k - 1$ binary classification tasks (Alcacer et al., 2024; Barbero-Gómez et al., 2021), each using a sigmoid activation function in the cross-entropy loss. However, the sigmoid function independently outputs the predictive probability for each category, meaning that the outputs are not mutually exclusive (Ledaguenel et al., 2024; Thaminkaew et al., 2024). This could lead to non-intuitive results even under the constraints of ordinal intervals. In contrast, the softmax function provides normalized probability outputs for mutually exclusive categories, facilitating a more natural handling of ordinal multiclassification.

3. Approach

3.1. Research objectives

In this work, we propose treating fashionability as a relative attribute to address the problem of comparing fashionability based on pairs of clothing images. To the best of our knowledge, this area has not been explored before. The objectives of this study are as follows:

- To develop a novel mathematical model for transforming the relative fashionability comparison problem, enabling more accurate and interpretable fashion ranking.
- To propose an effective method for overcoming the challenges posed by difficult-to-classify instances and ordinal attributes in the transformed fashionability comparison problem, improving ranking accuracy.

3.2. Problem formulation

This study reconceptualizes the comparative analysis of fashionability across ensemble imagery as an ordinal classification challenge. It involves annotating image pairs (m, n) with relative fashionability indices $S_{m,n}$, $S_{m,n} \in \{2, 1, 0\}$ as determined by experts in the field. We have:

$$S_{m,n} = \begin{cases} 2, S_m \succ S_n \\ 1, S_m \sim S_n \\ 0, S_m \prec S_n \end{cases} \quad (1)$$

Let us assume that the i -th image is represented in an input space $x_i \in \mathcal{X}$, and there is an outcome space $y_i \in \mathcal{Y} = \{r_1, r_2, \dots, r_K\}$ with ordered ranks $r_K \succ r_{K-1} \succ \dots \succ r_1$. The symbol $\succ$ denotes the ordering between different ranks. Given training samples $\mathcal{S}_{train} = \{x_i, y_i\}_{i=1}^{N_{train}}$, the objective of ordinal classification is to discover a function $h(\cdot) : \mathcal{X} \rightarrow \mathcal{Y}$ that maps images to their corresponding ranks, using a predefined cost $L : \mathcal{X} \times \mathcal{Y} \rightarrow \mathcal{R}$. The goal is to minimize the risk functional $\mathcal{R}(h)$.

In this paper, we utilize one of LightGBM's appealing features: the ability to supply a custom loss function (Gao et al., 2022; Sundaram & Tallamraju, 2021). Depending on the application, this capability can significantly enhance performance.

3.3. Our approach

Addressing fashionability comparisons between image pairs through dual-input network configurations is found to be complex and ineffective, based on initial experimental insights. Aligning with established pairwise ranking techniques (Aftab & Ramampiaro, 2024; Beutel et al., 2019), we pivot to a strategy of converting pairwise comparisons into a framework of pairwise difference learning (referenced in Fig. 2), thereby redefining the task as a conventional multi-class classification challenge: ordered pairs: Category 2 ($S_m \succ S_n$), Category 0 ($S_m \prec S_n$) and similar pairs: Category 1 ($S_m \sim S_n$).

Within this redefined framework, ordered pairs with visible differences in fashionability typically exhibit clear class separability in the feature space, making them easier to classify. In contrast, samples categorized as "similar" reside near the decision boundaries, where their discriminative features are relatively weak and prone to confusion with adjacent classes. The One-vs-Rest strategy (Guo

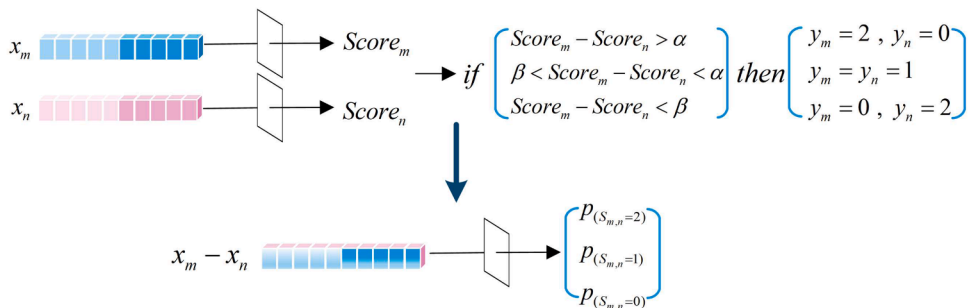


Fig. 2. Transform pairwise comparison to pairwise difference learning.

et al., 2020), which designates the samples of a single category as positive and amalgamates all other categories as negative, offers a partial remedy. However, the accuracy skew persists in outcomes for 1 vs. 0&2, attributable to the training dynamics for 0 vs. 1&2 and 2 vs. 0&1 configurations. This suggests that the model may encounter potentially difficult-to-classify and error-prone samples during the classification process.

To address difficult-to-classify samples, we choose to introduce the focal loss (Lin et al., 2020a; Liu et al., 2022) as an intervention. For the multiclassification scenario, we have:

$$L_{fl} = -\alpha_j y_{ij} (1 - p_{ij})^\gamma \log(p_{ij}) \quad (2)$$

wherein, a modulation factor, $(1 - p_{ij})^\gamma$, is introduced on top of the cross-entropy loss. By setting $\gamma > 0$, the relative loss of well-classified samples is reduced, thereby decreasing their gradient contribution during optimization and directing the overall gradient updates towards hard-to-classify examples.

Unlike difficult-to-classify samples, misclassified samples appear to be even more challenging to handle, as they involve a broader range of categories. Specifically, the fact that misclassified samples extend beyond similar pairs can be attributed to limitations in feature representation within the feature space. Even when ordered pairs exhibit pronounced differences, their differential features may be captured in a way that produces overlapping representations, which complicates the classification process. This phenomenon suggests that the issue arises not only from the inherent subjectivity in fashion, which blurs the mapping between labels and features, but also from challenges related to compactness in feature distance.

At this point, we propose further transforming the multiclassification problem into an ordinal multiclassification problem, in order to widen the feature distance between different categories. In fact, the ordinal relationship only exists between pairs of images. For example, when $m \succ n$, after the relative mapping transformation, $m = 2, n = 0$ indicates that the fashion level of m is higher than n . However, if we consider m as the baseline image, we can say that, according to the magnitude of the differences, there exists a kind of ordinal relationship among these image pairs (Fig. 3). This relationship reveals the order of the relative differences between image pairs, even though this order is derived from internal comparisons within each pair.

To address this ordinal multiclassification problem, we first examine the multiclass component. Unlike the sigmoid function, softmax solves the mutual exclusion of multiple classifications that cooperate with one-hot real label transformation (Fig. 4). When applied to the multiclassification problem, the difference of softmax classes probability is relatively significant, and the maximum probability value close to 1 means the model performed well. We assume ordinal multiclassification with K ranks, as $K > 2$, it is reasonable to employ softmax cross-entropy as the loss:

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^K y_{ij} \log(p_{ij}) \quad (3)$$

where y_{ij} and p_{ij} represent the label of the true category j of the i -th sample and the predicted probability, respectively.

Combined with ordinal information, we derived the custom objective function: ordinal focal loss as follows:

$$L_{ofl} = -\omega_j y_{ij} \alpha_j (1 - p_{ij})^\gamma \log(p_{ij}) \quad (4)$$

specifically, the ordinal focal loss inherits the basic structure of the cross-entropy loss, using $\log(p_{ij})$ to boost the predicted probability for the correct class; $(1 - p_{ij})^\gamma$ significantly amplifies the loss gradient for hard-to-classify samples, thus allocating more attention to them during training and actively "pushing" them toward the correct decision boundary. More importantly, this approach does not merely involve local weight adjustments, but rather induces a fundamental change in the distribution of samples within the global feature space, enabling the model to fundamentally enhance its ability to recognize challenging classes. Since this is a multi-class scenario, the role of α_j cannot be definitively determined at this stage. Meanwhile, ω_j , the ordinal weight for the j -th class, is introduced to further increase the model's sensitivity to the "distance" between classes. In our design, the distance between classes is not only reflected by the order of the labels but is also enforced in the feature space through ω_j , which imposes a stronger penalty on misclassifications that deviate further from the correct class. Thus, the overall loss, propagated backward, optimizes based on the relative distances between classes, gradually enabling the model to "learn" to construct decision boundaries in the feature space that are more sensitive to inter-class differences.

We visually explore the effect of adjusting the parameter ω on our custom loss function through simple graphics (see Fig. 5). As ω increases, the loss for a given prediction probability y_{pred} also increases, reflecting a more stringent penalty for classification errors. By adjusting ω for each category, we can flexibly embed ordinal information in multi-classification models.

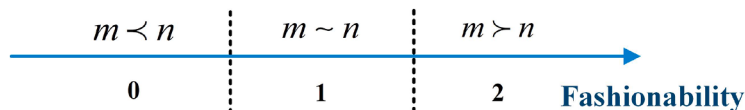


Fig. 3. Ordinal pairwise difference learning.

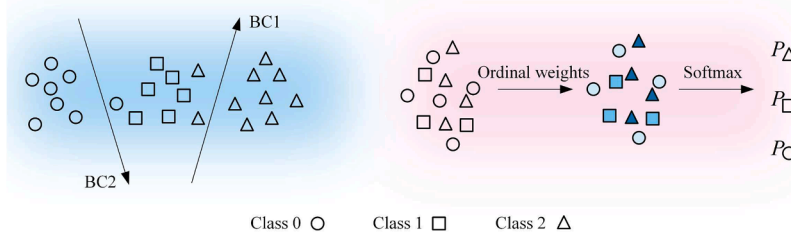


Fig. 4. To address K – level ordinal multi-class problems, it is common to transform them into $K - 1$ binary classification problems. Binary Classifier 1 (BC1) distinguishes class 2 from classes 0&1, while Binary Classifier 2 (BC2) separates classes 2&1 from class 0. A prediction of class 1 is made when BC1 is false and BC2 is true. This method preserves ordinal information well but requires high-class exclusivity to prevent overlap (as depicted in the blue background). Fashion levels, for instance, are not absolutely mutually exclusive, an image might be more fashionable compared to another but similar to a third. Handling this as a multi-class task with ordinal weights between classes (as in the pink background) can enhance differentiation and ensure exclusive outputs.

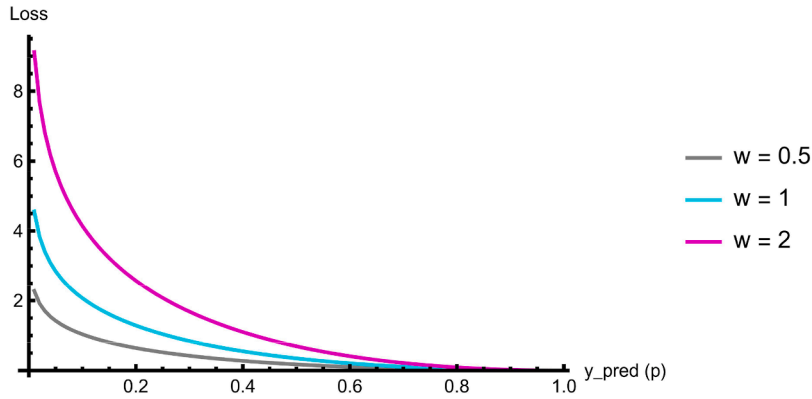


Fig. 5. Impact of ω on loss function.

3.4. Learning the custom objective function

To successfully implement a custom objective function for LightGBM, there are four steps that need to be taken.

- Custom objective function and metric, which we have completed.
- Derive the first and second derivatives of the objective function with respect to the predicted values z_j to update the gradient and Hessian matrices (the detailed derivation process is too complex to be omitted here, see [Appendix A](#)). Here, according to the chain rule, we have first-order derivative:

$$\frac{\partial L_{off}}{\partial z_j} = \omega_j y_i \frac{\partial L_{fl}}{\partial p_i} \frac{\partial p_i}{\partial z_j} \quad (5)$$

$$\frac{\partial L_{fl}}{\partial p_i} \frac{\partial p_i}{\partial z_j} = \begin{cases} \alpha_i [t \gamma p_i - (1 - p_i)^{\gamma+1}], & \text{if } i = j \\ -\alpha_j p_j \left[\frac{t \gamma p_i}{1 - p_i} - (1 - p_i)^{\gamma} \right], & \text{if } i \neq j \end{cases} \quad (6)$$

For second order derivative, we construct the following chain derivation steps:

$$\frac{\partial^2 L_{off}}{\partial z_j^2} = \omega_j y_i \left(\frac{\partial}{\partial z_j} \left(\frac{\partial L_{fl}}{\partial p_i} \right) \frac{\partial p_i}{\partial z_j} + \frac{\partial L_{fl}}{\partial p_i} \frac{\partial}{\partial z_j} \left(\frac{\partial p_i}{\partial z_j} \right) \right) \quad (7)$$

Among them, let: $\varepsilon = \frac{\partial}{\partial z_j} \left(\frac{\partial L_{fl}}{\partial p_i} \right) \frac{\partial p_i}{\partial z_j} + \frac{\partial L_{fl}}{\partial p_i} \frac{\partial}{\partial z_j} \left(\frac{\partial p_i}{\partial z_j} \right)$,

$$\varepsilon = \begin{cases} \alpha_i p_i (1 - p_i) \left(-\frac{t\gamma^2 p_i}{1 - p_i} + \gamma t + (2\gamma + 1)(1 - p_i)^\gamma \right), & \text{if } i = j \\ \alpha_j \left(p_i p_j^2 \left[\frac{t(\gamma - \gamma^2) p_i}{(1 - p_i)^2} + \frac{t\gamma}{1 - p_i} + 2\gamma(1 - p_i)^{\gamma-1} \right] - p_j (1 - p_j) \left[\frac{\gamma p_i}{1 - p_i} - (1 - p_i)^\gamma \right] \right), & \text{if } i \neq j \end{cases} \quad (8)$$

- Define an initialization value for the training and validation set. In the pseudocode for gradient boosting algorithms, it's generally shown that the model starts from an initial estimate, and an initial value δ is set for faster convergence. The definition of δ depends on the loss function. By default, LightGBM defaults to $\delta = 0$, which is sub-optimal. It can be seen from the expression of the initial estimate that the initial value that minimizes the objective function is optimal.
- Add the initialization value to the test margins before converting them to probabilities.

The pseudocode of our approach for solving ordinal multiclassification with the focal loss is shown in [Algorithm 1](#).

4. Experiments

To confirm its effectiveness, we conducted comprehensive experiments to answer the following research questions: (1) How does proposed ordinal focal loss perform compared with other methods? (2) How do visual features and hyper-parameters affect performance? (3) How consistent are the model test results with human subjective recognition?

4.1. Datasets

Experiments on three datasets were conducted to evaluate the performance of the proposed method: our self-constructed Fashionability3k and two mainstream fashion datasets, Polyvore ([Han et al., 2017](#)) and EVALUATION3 ([Zou et al., 2020](#)). They all meet the following basic requirements for the data: (1) global images of outfits, (2) local images of items within the outfit, (3) textual descriptions of each item, and (4) relative fashionability labels. The attributes of the three datasets are listed in [Table 2](#).

4.1.1. Fashionability3k

Although several outfit datasets (multiple items as an outfit in one image) have been made publicly available for fashion compatibility modeling, such as Polyvore409k ([Tangseng et al., 2017](#)), IQON3000 ([Song et al., 2019](#)) and ModAI ([Balim & Özkan, 2023](#)), they are either plagued by noisy backgrounds or overloaded with accessories (such as bags, necklaces, earrings, etc.), which make the inherently subjective task of fashion assessment even more challenging. Furthermore, this study intends to achieve fashion ranking in clothing ensemble images through relative attributes. Therefore, annotating comparative fashion relationships among images becomes particularly crucial. In summary, we introduce a new Fashionability3k dataset (each outfit includes clothes and shoes but not accessories), specifically targeting the global attribute (fashionability) comparison task based on the online fashion-oriented community website SHOPLOOK¹.

Each outfit sample, or “Feed,” in SHOPLOOK’s terminology, consists of a composed visual image (showing the position of each item in the outfit), an image of each item, and a textual description for each item (including type, materials, pattern, etc.). Through tracking ‘similar outfits’ and ‘style it’, we can obtain more. It is worth noting that the number of likes on outfits in the SHOPLOOK is not directly related to their fashionability, as some high-liked outfit images highlight elements such as celebrity posters or makeup rather than the clothing itself. [Fig. 6](#) shows a SHOPLOOK feed and items.

Data cleansing: To ensure that the fashionability annotations in our constructed dataset are not influenced by external factors beyond the outfit composition itself—primarily including multiple or incomplete outfits, extraneous elements in the outfit visual images (such as decorative backgrounds and accessories), and subjective terms in the outfit textual descriptions—we employ a three-step processing procedure. First, remove samples that contain more than one outfit and those that are insufficient to form a complete outfit. Following the work of [Tangseng et al. \(2017\)](#), [Lin et al. \(2020b\)](#), this study defines an outfit as complete if it comprises the essential clothing elements that cover the body from top to bottom; Second, cleaning outfit visual images by removing decorative backgrounds and accessories using Photoshop’s lasso tool. In the initial stage, we attempted batch processing using the pre-trained SAM model ([Kirillov et al., 2023](#)). However, due to the complexity of the decorative backgrounds, the entire process became more complex and time-consuming. The removed areas are filled with white (RGB: 255, 255, 255) to avoid affecting subsequent feature extraction; Lastly, text description cleaning: for the textual descriptions of items included in each outfit, in order to ensure accurate filtering, we manually review and remove subjective terms (as defined by [Yang et al., 2020](#)), brand names, prices, and other interfering information (e.g., special characters), since automated tools may not fully capture the complexity of domain-specific terminology and contextual nuances. After filtering, we obtained 3500 outfits, each containing an arbitrary number of items ranging from 2 to 5, with an average of 3 items per outfit (see [Fig. 7](#)).

Data labeling: We need paired data that has already been labeled with relative fashionability to learn a global ranking function for evaluating the fashion strength of new outfits. During this process, two points require special attention. First, we prefer that the

¹ <https://shoplook.io>.

Algorithm 1

Ordinal focal loss training strategy.

Define $L_{off}(N, K, \alpha, \gamma, \omega, Z, Y)$:**Input:** $N, K, \alpha, \gamma, \omega$ are as defined before. Z : The raw output matrix of the model, where each element z_{ij} represents the raw prediction score of sample i for class j . Y : True label matrix. The element y_{ij} is 1 if sample i belongs to class j and 0 otherwise.**Output:** L_{off} , Gradients (G), Hessians (H)**Start:****Initialize**res: = minimize (L_{off} , initial_logits)

Return res.x

For each sample i from 1 to N :**For** each category j from 1 to K : $z_{ij} = Z[i, j]$, $p_{ij} = \text{softmax}(z_{ij})$, $y_{ij} = Y[i, j]$, $\alpha_j = \alpha[j]$, $\omega_j = \omega[j]$ $L_{off} = \text{Eq. (4)}$ $G[i, j] = \omega_j y_{ij} \text{Eq. (6)}$ $H[i, j] = \omega_j y_{ij} \text{Eq. (8)}$ **Return** L_{off}, G, H **Define train model with** L_{off} ($\mathcal{S}_{train}$, $\mathcal{S}_{valid}$, $\mathcal{S}_{test}$, Θ):**Input:** $\mathcal{S}_{train}$, $\mathcal{S}_{valid}$, $\mathcal{S}_{test}$; $\mathcal{S} = \{x_i, y_i\}_{i=1}^N$ Θ : Learning rate (η), number of iterations ($iter$), α , γ , ω

init_value: Initialization logits preset by optimization.

Process:1. Initialize the model, configuring L_{off} as the objective function.

2. For each iteration:

a. $Z_{ij} = f(x_i; \Theta)_j$ for $i = 1, 2, \dots, N_{train}$ b. $(L, G, H) = L_{off}(Z, Y; \alpha, \gamma, \omega)$ c. $\Theta \leftarrow \text{Update}(\Theta, G, H)$ d. $\Theta_{opt} \leftarrow \text{Optimize}(\Theta, \mathcal{S}_{valid}, \text{Metric}(\text{Evaluate}(\mathcal{S}_{valid}, \Theta)))$ 3. $Z_{ij} = f(x_i; \Theta_{opt})_j + \text{init_value}$ for $i = 1, 2, \dots, N_{test}$ 4. $p_{ij} = \text{softmax}(Z_{ij})$ for $i = 1, 2, \dots, N_{test}$ **Output:** $p = \{p_i\}_{i=1}^{N_{test}}$ **Table 2**

The statistics of the Fashionability3k, Polyvore and EVALUATION3 datasets.

Dataset	Outfit	Items in outfit	Fashionability labels
Fashionability3k	3000	Variable, 2 to 5	Relative value (more; similar; less)
Polyvore	7076	Variable, 2 to 8	Absolute value (Compatible; Incompatible)
EVALUATION3	16,371	4	Absolute value (good; normal; bad)

**Fig. 6.** Example of SHOPLOOK outfits.

samples being compared are roughly at the same level, rather than samples with vastly different fashionability. This aligns with the perspective emphasized in the introduction—that consumers often deliberate between outfits of similar fashion quality. Second, a perfectly ordered pairwise list would require the individual to perform a large number of comparisons, especially since our sample size will need more than 4 million times to get all possible pairwise image combinations. To find a balance between efficiency and effectiveness, this paper draws on the ten-bin method in [Altwayjry and Belongie \(2013\)](#), which can roughly categorize samples by



Fig. 7. Example of Fashionability3k dataset.

fashionability level and significantly reduce the number of required comparisons.

We first invited a fashion expert (different from those who will annotating fashionability later) to perform a rough fashionability binning. We acknowledge that inviting more fashion experts would enhance credibility. However, the decision to rely on a single expert for initial fashionability assessments is justified by the following considerations: (1) The selected expert possesses extensive industry experience and specialized knowledge, capable of providing valuable preliminary fashionability assessments. (2) Due to the subjective nature of fashionability assessments, aggregating evaluations from multiple experts would be time-consuming, underscoring the importance of consistency in assessment. (3) The primary aim of the initial fashionability assessments is to avoid substantial disparities among samples, with subsequent focus on pairwise comparisons (involving 10 experts for annotation). Therefore, relying on a single expert's assessments adequately meets our research needs and does not significantly impact the final results.

This process was carried out using the simple binned-sort interface we designed. Specifically, 10 bins are being used to emulate fashion levels 1 to 10, where we have $bin_{10} > bin_9 > \dots > bin_2 > bin_1$.

Subsequently, we further invited fashion experts to use a GUI designed for comparing fashionability. For each pair, experts were required to make a relative decision (i.e., more, less or equal) and report the confidence of their decision (high, medium, or low). During the comparison process, a simplified comparison method (comparing only adjacent samples) was employed, starting with the first and second samples from bin 10, and continuing until all samples across the 10 bins had been pairwise compared. By default, the last sample in a higher bin was considered more fashionable than the first sample in a lower bin, thereby facilitating cross-bin comparisons; in such cases, no additional expert intervention was necessary. In total, 10 fashion experts participated in the entire process, and the final ground truth was determined through a voting procedure. The entire data labeling process is shown in Fig. 8. After removing the low confidence (average confidence score > 2) or agreement pairs (agreement score < 0.5), we are left with 2999 total pairs (601 similar pairs and 2398 ordered pairs).

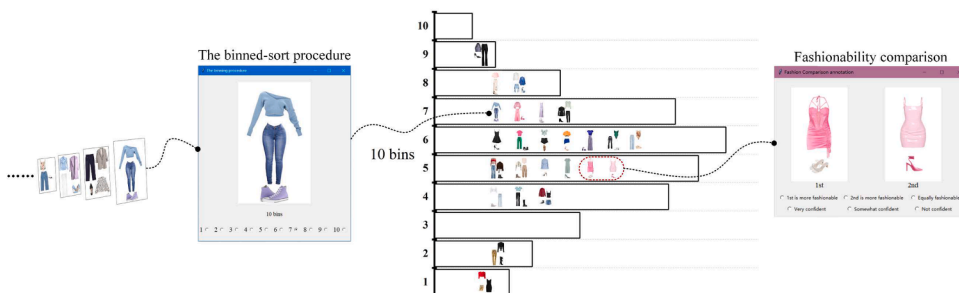


Fig. 8. Relative fashionability annotation.

4.1.2. Other datasets

Different datasets typically have varying characteristics. To avoid potential biases and specificities inherent in the self-constructed dataset, we also conducted experiments on two additional mainstream datasets to evaluate the performance of the ordinal focal loss when confronted with these diverse challenges.

Polyvore (Han et al., 2017) contains a total of 7076 outfits with labels indicating whether they are matching, where 3076 are compatible (labeled as 1) and 4000 are incompatible (labeled as 0). By generating non-redundant random pairs and converting them into relative fashionability labels according to Eq. (1), we can establish a fashionability comparison relationship for any pair of outfits. Among the final 3538 pairs, there are 1710 ordered pairs and 1828 similar pairs.

EVALUATION3 (Zou et al., 2020) provides fashionability labels with three levels: good (labeled as 1), normal (labeled as 2), and bad (labeled as 3). Among these, only the E-VC and F-Test3000 subsets meet the requirements, comprising a total of 16,371 outfits. Using the same fashionability comparison conversion method as in the Polyvore dataset, 8185 pairs are ultimately generated, including 2969 ordered pairs and 5216 similar pairs.

4.2. Implementation

Feature representation: Lin et al. (2019)'s experiments have found that using complete images of fashion outfits yields better results when learning an individual's taste in fashion. However, existing studies overlook the global features of the outfit (Fig. 9(a)) and instead represent an outfit by sequentially extracting and concatenating the visual features of its individual items (Fig. 9(b)). To address this, we propose a method that combines global and local features for outfit representation, as shown in Fig. 9(c).

First, the focus is on the outfit's global and textual features. Research dedicated to fashion recommendation and compatibility modeling often employs different models to extract image and text features separately (Sarkar et al., 2023; Wang et al., 2023). However, these models more or less overlook the correspondence between images and text because they are trained independently. Splicing such extracted multimodal features usually requires some feature engineering before they can be used as training data. Especially in the fashion domain, text often emphasizes the detailed descriptions of different products, which is crucial for comparison tasks between images. We need a strong correspondence between multimodal features so that they can cooperatively reflect similarities and also highlight unique advantages; otherwise, text features will be redundant.

CLIP model (Radford et al., 2021) has shown a superior capability on word/region token level alignment to other methods (Li et al., 2024), which is exactly what we need. We start from the released pre-trained CLIP model, which applies ViT-B (Dosovitskiy et al., 2021) as the visual encoder $x_{global}(\cdot)$ and the Transformer (Vaswani et al., 2017) as the text encoder $x_{text}(\cdot)$. The input image is resized to 224×224 , and the input text is tokenized using a lower-cased byte pair encoding method. We finetune the pre-trained CLIP model, following Gao et al. (2020), Zhuge et al. (2021), who delved into fashion field data. The model encodes image and text into 512 dimensions, respectively, as feature representation. We randomly presented the cosine similarity results of image-text features for 8 data points (Fig. 10). In addition to accurately reflecting the correspondence between the outfits and their corresponding text descriptions, we were also pleasantly surprised to find that for outfits of similar styles (2&3, 4&7), the model gave almost identical similarity scores, whereas for completely different styles (for example, 8), the scores were significantly higher than the others.

For local features, to leverage the advantages of the Transformer architecture while achieving efficient computation under resource constraints (with each outfit containing between 2 and 8 local items, leading to a significant increase in computational load), ViT-small (Dosovitskiy et al., 2021) was selected. To match the dimensionality of local features with global features, each item is set to a dimensionality of 384. When an outfit has fewer items than the maximum, we pad with the average features of the existing items. After extraction, the preliminary local feature dimensions are 1920-D, 3072-D, and 1536-D for the three datasets. Following linear dimensionality reduction, the local features $x_{local}(\cdot)$ are standardized to 512-D.

Thus, the input feature x satisfies the following expression:

$$x = (x_{global} \oplus x_{text}) \parallel x_{local} \quad (9)$$

Experiment settings: We utilize stratified sampling to divide the dataset into 80 % for training, 10 % for validation, and 10 % for testing (see Table 3 for details). For all comparison methods, we employ identical training, validation, and test splits.

We report F1-score in terms of the harmonic mean of precision and recall for correctly classified pairs, following Yu and Grauman (2014). Our model optimization is based on the calculation of gradients and the Hessian matrix of a custom objective function, which achieved optimal performance with Num_leaves is 60, Max_depth reaches 9, and the hyperparameters γ , α , ω are set as 2, [0.5, 0.5, 0.5]

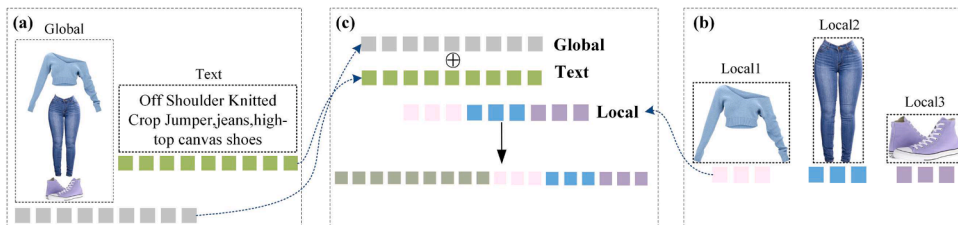


Fig. 9. Proposed feature representation method.

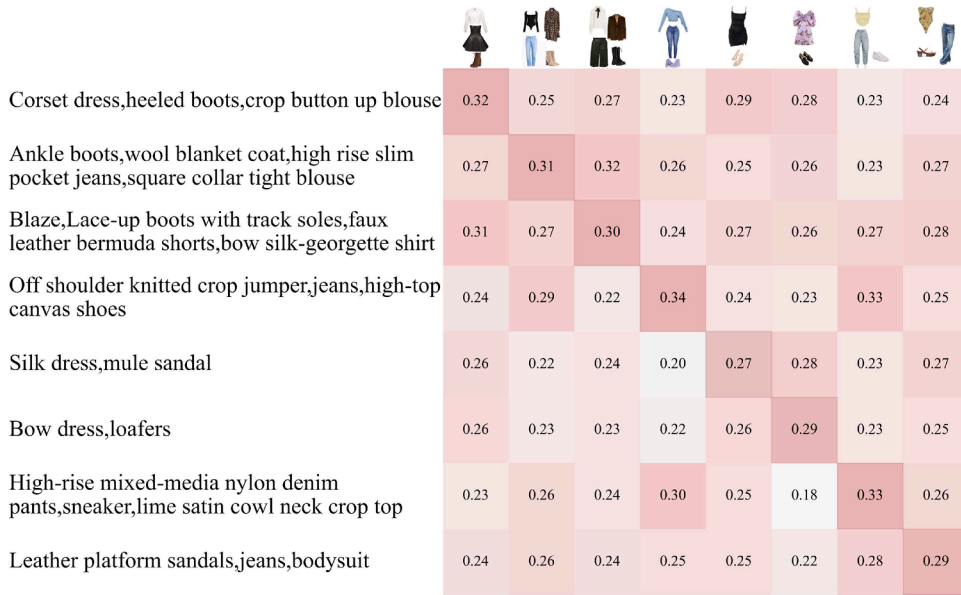


Fig. 10. Cosine similarity between text and image features.

Table 3

Dataset partitioning details.

Dataset	Train	Validation	Test
Fashionability3k	2429	270	300
Polyvore	2865	319	354
EVALUATION3	6629	737	819

and [0.1, 0.3, 0.6], respectively. All experiments are conducted on a single NVIDIA GeForce RTX 3070Ti GPU.

4.3. Model performance (RQ1)

Convergence analysis: Fig. 11 visualizes the convergence curves of the loss function for three datasets. For the Fashionability3k dataset, which was specifically constructed to address comparative fashionability, there is a high degree of consistency between the dataset and the custom objective function. The curve exhibits a low initial loss, and as the number of iterations increases, the loss decreases steadily and rapidly, confirming the efficiency of the objective function on this dataset. In contrast, the Polyvore and Evaluation3 datasets show higher initial loss values and slower rates of decrease, indicating that the model's initial fit on these datasets is less optimal. However, despite the relatively lower consistency with the objective function, the loss values gradually decrease with more iterations, demonstrating that the model can progressively learn and optimize. The preliminary fit analysis confirms the generalization capability of the objective function across different datasets.

Comparison with baselines: Since we are the first to introduce the comparison of fashionability as a relative attribute, we selected methods that are both reasonable and effective under the context of this research problem to address the issue and achieve the research objectives. First, as the relative comparison of fashionability is essentially a pairwise ranking problem, we chose to compare our approach with RankSVM (Joachims, 2002; Meng et al., 2024), which is widely used for solving relative ranking problems. Second, this study highlights the limitations of viewing fashionability comparison as an absolute binary classification problem, as fashion is inherently relative. Therefore, we selected the Global method proposed by Parikh and Grauman (2011) for comparison, in order to demonstrate the advantage of applying relative attributes in our method. Next, following our preliminary results and in line with recent pairwise ranking research (Aftab & Ramampiaro, 2024), we transformed pairwise ranking into a multi-class classification problem based on pairwise differences. We also found that differences from relative comparisons produce ordinal attributes. Incorporating these ordinal attributes into the classification model can enhance its ability to distinguish between misclassified samples, thus widening their classification boundaries.

Given the high dimensionality, data volume, and preliminary results, we consider LightGBM an efficient framework for this study. LightGBM has been shown to be effective in multiple studies (Bin Tareaf et al., 2024; Deng et al., 2024; Harada et al., 2024; Liu & Hu, 2024; Nguyen et al., 2024) and often outperforms neural network-based methods in terms of performance and robustness (Nguyen et al., 2024; Sajid et al., 2024). Therefore, this study also compares Focal loss (Guo et al., 2020; Lin et al., 2020a) and LightGBM's Multi logloss as baselines.

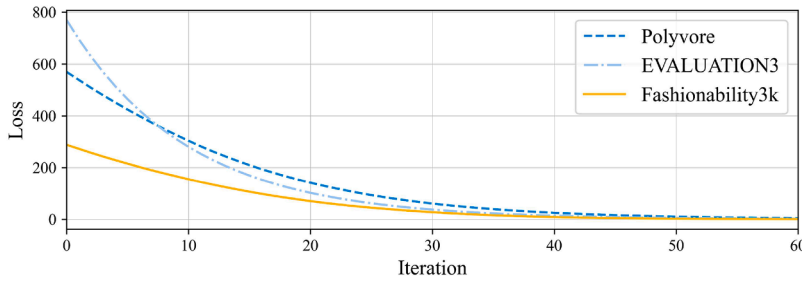


Fig. 11. Convergence curves of the custom loss function.

These baselines are essential for evaluating our method and offer reliable comparison points. All methods were tested on the same three datasets, with results independently run and reported by us. A brief overview of the baselines is as follows:

- RankSVM: A method initially proposed to transform ranking problems into pairwise comparison issues (Joachims, 2002). It aims to maximize the boundary margin between ordered pairs.
- Global: A global ranker trained with all available labeled pairs. This is the Relative Attributes Method (Parikh & Grauman, 2011), which based on the RankSVM framework, using hinge loss to measure ranking errors and thereby transforming the ranking problem into an optimization problem with margin constraints. The original code given by the authors was in MATLAB, so we used the Python version in GitHub².
- LightGBM-multi logloss: An efficient gradient boosting algorithm based on decision trees (Ke et al., 2017), which can be employed to solve classification and regression problems. In LightGBM, the multi-class logarithmic loss is the default loss function used for multi-class tasks and does not consider ordinal information. Through comparison, we aim to verify whether the customized strategy is as effective as anticipated.
- LightGBM-focal loss (Guo et al., 2020): In our task, there exist hard-to-classify samples. Therefore, without initially considering ordinal information, we adjust the focus parameter to assign higher weights to the hard-to-classify and easily misclassified samples, in order to observe the performance.
- LightGBM-ordinal focal loss: The proposed ordinal multi-classification approach.

The F1-score (Table 4) and the PR curve in Fig. 12 (which only presents results on the Fashionability3k dataset) collectively depict a comprehensive performance evaluation, leading to the following observations:

- Overall, the proposed method demonstrates superior and stable performance. To validate the advantages of the relative attribute theory in addressing such problems, we categorize methods into two groups: non-relative attribute group (RankSVM, Lgb-multi_log and Lgb-focal) and relative attribute-based group (Global and Lgb-ordinal focal) for comparison. Cross-group comparisons reveal that methods utilizing relative attributes generally achieve superior and more stable performance in classification accuracy. Particularly, they exhibit improvements of approximately 2.58 % in EVALUATION3 and 0.88 % in Fashionability3k for similar category judgments, respectively.
- RankSVM exhibits a unique advantage in similar pair evaluation, particularly on Fashionability3k, where it achieves results nearly on par with the proposed method. Its continuous optimization of decision boundaries helps to better capture subtle differences between samples. However, RankSVM shows certain limitations in ordered pair evaluation, especially on Polyvore and EVALUATION3. Additionally, its performance varies significantly across different datasets, indicating weaker cross-dataset adaptability.
- In contrast, the Global performs exceptionally well in ordered pair evaluation, achieving the highest score on EVALUATION3. The global method, based on nonlinear learning, is ideal for tasks with clear sequential relationships. However, this method is a general form of relative attribute learning that typically starts from the perspective of distinguishing which image is stronger in a specific attribute, but it does not specialize in subjective and fine-grained attributes like "fashionability". In contrast, our method clearly incorporates improvements highly relevant to the fashion domain, such as refined feature extraction, label design, or loss functions (e.g., introducing ordinal relationships, focal loss), making it more adaptable to discern subtle differences in clothing coordination and fashionability judgments.
- The multi_logloss and the focal loss are balanced overall, with no standout strengths in any one area but consistent performance across tasks. Focal loss shows strong stability, especially on EVALUATION3 and Fashionability3k, and handles multi-task requirements well. Multi_logloss lags slightly behind, particularly in the similarity task on Fashionability3k, suggesting it struggles with more complex or varied data.

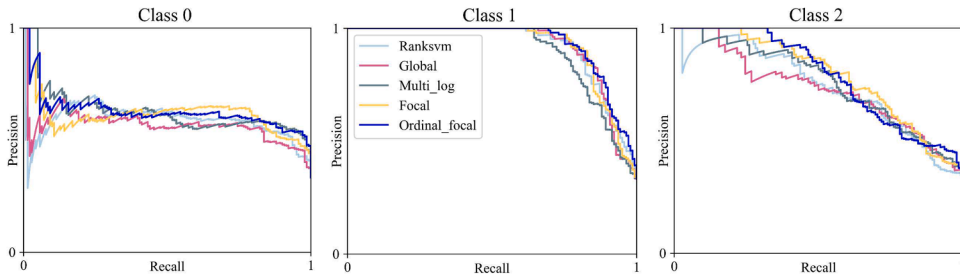
To further investigate the performance of our proposed method in classification tasks (inter-class misclassification relationship), we compared the confusion matrices under three different loss conditions. We believe that confusion matrices have limited readability in

² <https://github.com/chaitanya100100/Relative-Attributes-Zero-Shot-Learning>.

Table 4

State-of-the-art F1-score (%) compared with the results obtained in this work. Bold figures indicate the best results.

Dataset	Approach	Ordered	Similar	Overall
Polyvore	RankSVM	75.54	66.82	72.13
	Global	79.89	60.57	73.77
	Lgb-multi_log	80.31	65.57	75.41
	Lgb-focal	79.02	65.13	74.13
	Lgb-ordinal focal(Ours)	81.79	69.04	77.23
EVALUATION3	RankSVM	83.02	76.95	80.64
	Global	89.04	80.43	86.20
	Lgb-multi_log	87.66	79.19	84.73
	Lgb-focal	87.74	80.14	85.05
	Lgb-ordinal focal(Ours)	88.47	82.24	86.26
Fashionability3k	RankSVM	65.64	85.17	71.70
	Global	64.40	80.82	70.05
	Lgb-multi_log	63.36	78.64	68.63
	Lgb-focal	68.00	83.21	72.88
	Lgb-ordinal focal(Ours)	68.09	85.61	73.82

**Fig. 12.** Precision-recall curve on Fashionability3k dataset.

analyzing the classification capability of models; therefore, we transformed them into "distance features" and utilized Multidimensional Scaling (MDS) techniques for mapping the relative distances between categories. As evident from Fig. 13, the performance of multi_logloss across the three datasets is relatively basic, primarily relying on the independence of each category for classification. However, there is a significant overlap between the categories in all cases. Focal loss achieved a certain degree of improvement. In the Polyvore and EVALUATION3 datasets, the normalized distance between Class 0 and Class 1 remains relatively short, indicating a high confusion rate between them, though it has improved compared to multi_logloss. Particularly in the Polyvore dataset, the normalized distance with focal loss has increased, leading to a slight improvement in inter-class separability. However, the distinction between Class 2 and the other classes is not significant, especially in the relatively complex Fashionability3k dataset, where class overlap remains prominent, revealing certain limitations of the model in handling inter-class relationships. Our method enhances classification accuracy by effectively reducing inter-class misclassification. It maintains strong separability between classes across the three datasets, particularly among categories with similar features. The introduction of ordinal relationships enables our method to not only focus on the independence of categories but also to gain a deeper understanding of the relationships between them, thereby significantly improving the model's generalization performance. This is reflected in the figures as clearer class boundaries and reduced inter-class overlap.

4.4. Discussion of visual features and hyper-parameters (RQ2)

Visual feature analysis: To translate the theoretical benefits of global features in fashionability evaluation into practical results, we conducted additional comparative experiments. The results, shown in Table 5 (excluding text features for fairness), indicate that the local and global visual feature combination significantly boosts performance further, with average gains of 2.78 % in ordered pairs and 1.09 % in similar pairs. This suggests that fashion evaluation relies not only on individual item features but also on the coordination and overall style of the outfit. While local features capture item-specific details crucial for identifying similar pairs, they fail to account for the overall style formed by the combination.

From a methodological perspective, we integrated textual features and plotted the overall metric trends for the relative fashionability comparison task with three feature inputs, as shown in Fig. 14. Our method shows positive sensitivity to feature combinations, with performance improving as the feature set expands. This effective use of multi-dimensional data highlights the method's strong adaptability and stability.

Parameter sensitivity evaluation: This section provides a detailed analysis of the impact of the model's ordinal weights ω_j and the hyperparameters α_j and γ on the model's performance. Given that the rank order in this study is not absolute, we adopt the definition of varying class values from weighted entropy (Singer et al., 2020) to assign values to the ordinal weights. The hyperparameters α_j , γ

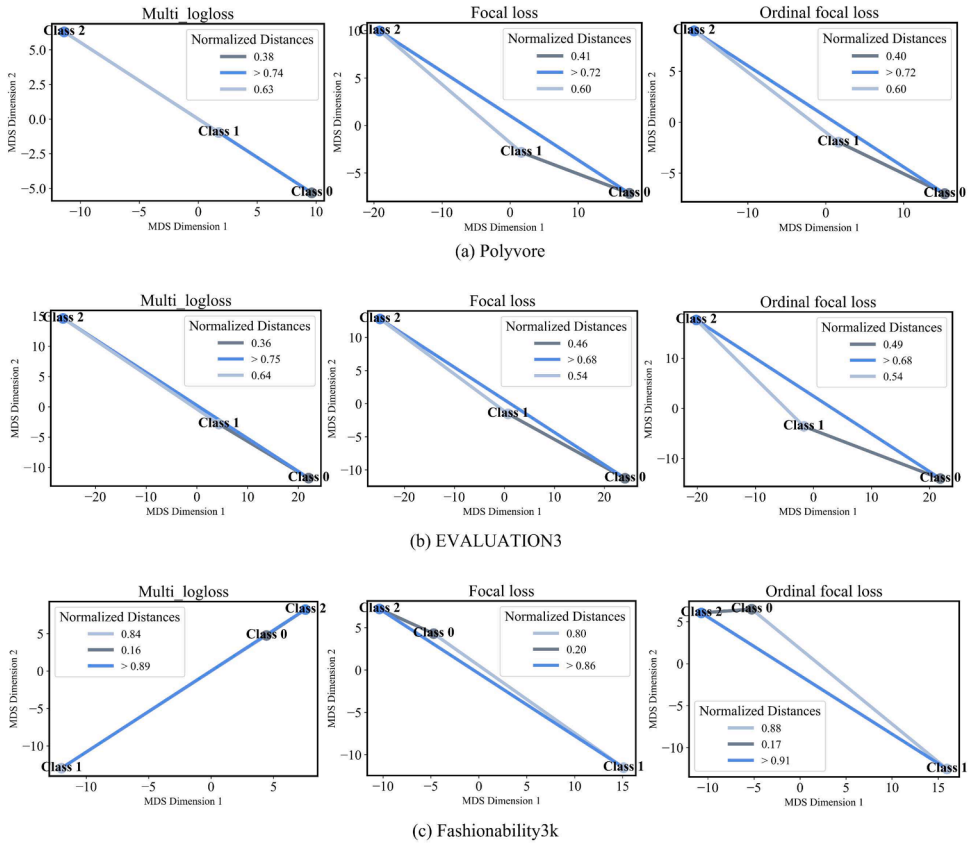


Fig. 13. Two-dimensional scaling visualization of confusion matrices for different loss functions on three datasets.

Table 5

F1-score (%) comparison before and after adding global information to visual features ($\uparrow$ indicates improvement).

Dataset	Approach	Local			Local-global			$\Delta_{overall}$
		Ordered	Similar	Overall	Ordered	Similar	Overall	
Polyvore	RankSVM	74.12	66.36	71.04	74.69	66.36	71.40	$\uparrow$ 0.36
	Global	76.75	55.42	70.31	79.99	60.34	73.77	$\uparrow$ 3.46
	Lgb-multi_log	73.52	60.16	69.03	78.80	62.30	73.04	$\uparrow$ 4.01
	Lgb-focal	76.05	65.05	71.95	79.00	60.96	72.86	$\uparrow$ 0.91
	Lgb-ordinal focal(Ours)	77.71	64.12	72.86	79.84	64.29	74.32	$\uparrow$ 1.46
EVALUATION3	RankSVM	82.73	76.44	80.26	82.74	76.72	80.38	$\uparrow$ 0.12
	Global	87.47	77.21	84.09	89.78	80.32	86.71	$\uparrow$ 2.62
	Lgb-multi_log	85.87	75.69	82.43	87.89	79.81	85.11	$\uparrow$ 2.68
	Lgb-focal	86.70	79.82	84.22	88.02	80.90	85.50	$\uparrow$ 1.28
	Lgb-ordinal focal(Ours)	87.76	81.79	85.62	88.73	81.95	86.33	$\uparrow$ 0.71
Fashionability3k	RankSVM	63.82	83.21	69.81	64.48	84.53	70.75	$\uparrow$ 0.94
	Global	55.43	80.00	64.15	64.20	82.07	70.28	$\uparrow$ 6.13
	Lgb-multi_log	56.18	76.35	63.21	60.17	78.32	66.27	$\uparrow$ 3.06
	Lgb-focal	60.66	80.43	67.22	62.26	83.33	69.10	$\uparrow$ 1.88
	Lgb-ordinal focal(Ours)	62.51	86.55	70.28	68.46	82.73	73.11	$\uparrow$ 2.83

respectively determine the model's balance between positive and negative samples and its focus on hard-to-classify samples. We assign values to α for the third category that specifically increase focus on positive samples, keeping the attention to positive and negative samples of the other two categories unchanged. Since the effect of α may not be as pronounced in multi-class problems as it is in binary tasks, we also set α to 0.5 for all three categories to observe the outcomes. Lastly, we set γ starting from 0.2 and increasing to observe the degree of attention to hard-to-classify samples. Specific parameter settings are shown in Table 6.

Since the Fashionability3k dataset is more aligned with the model's objective, this study conducts hyperparameter analysis on this dataset to ensure the relevance of the analysis and the reliability of the experimental results. As shown in Fig. 15(a), increasing values of γ positively impact both accuracy and loss, but excessively high values deteriorate the performance. The model achieves optimal

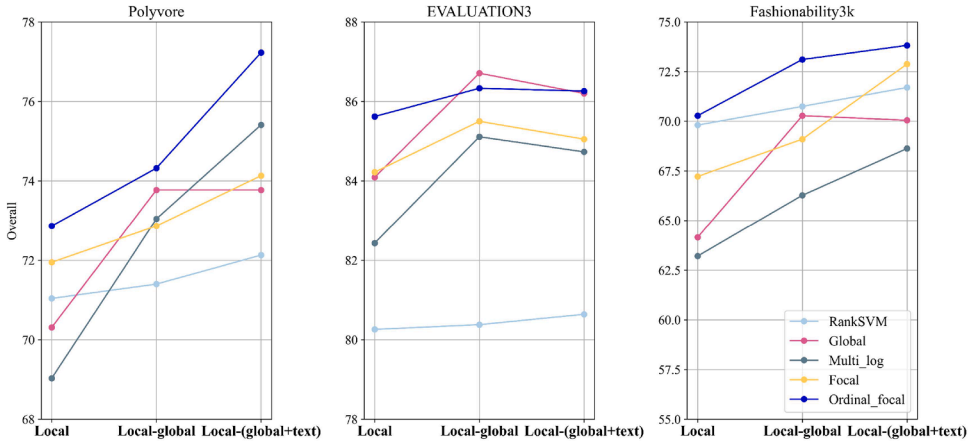


Fig. 14. Overall metric trend distribution for relative fashionability comparison across different feature combinations.

Table 6
Specific settings of hyperparameters.

Parameter	Value
γ	0.2, 0.3, 0.5, 0.7, 1, 2, 4
α_j	$\alpha_1 : [0.5, 0.5, 0.75]$, $\alpha_2 : [0.5, 0.75, 0.5]$, $\alpha_3 : [0.75, 0.5, 0.5]$, $\alpha_4 : [0.5, 0.5, 0.5]$
ω_j	$\omega_1 : [0.401, 0.066, 0.533]$, $\omega_2 : [0.1, 0.8, 0.1]$, $\omega_3 : [0.1, 0.3, 0.6]$

performance when γ is set to 2, which provides an initial assessment of the difficulty level of classifying samples. Subsequently, by fixing γ and combining different values of α and ω , we determine the best combination, as illustrated in Fig. 15(b). The results indicate that the highest accuracy is achieved when $\alpha = \alpha_4$ and $\omega = \omega_3$, as denoted by a pink pengagram. We understand that using weighted entropy for ordinal weight calculation can help the model more effectively distinguish and prioritize those categories that are rarer or of higher ordinal importance in the dataset. The optimal values of ω demonstrate the ordinal nature of the fashion trend comparison task and the positive impact of integrating ordinal information on classification. However, the optimal values of α seem to suggest that the positive and negative adjustment factors have a minimal impact on multi-class tasks. To corroborate this, we separately recorded their respective impacts on accuracy, as shown in Fig. 15(c). The results, as depicted, indicate that the bias towards either side of α does not significantly affect the outcome.

4.5. Human approval test (RQ3)

In order to evaluate the compatibility of the approach above with human subjective perception, inspired by Alshan et al. (2018), Altwaijry and Belongie (2013), Zou et al. (2022)2022), we took an extra step. If the experimental data from this phase is returned to the fashion experts who were involved in the data annotation for reassessment, they might be subject to the 'anchoring effect,' meaning they could rely on or be biased towards their initial choices. Therefore, we recruited 35 new fashion experts and presented them with randomly selected pairs of images, comprising 10 % of the test set, to record their majority voting as a measure for human evaluation of comparative fashionability.

We employ Cohen's Weighted Kappa ($\mathcal{K}$) (Yilmaz & Demirhan, 2023), suitable for ordinal classification, to assess the consistency between algorithm predictions and human evaluations.

$$\mathcal{K} = \frac{Po - Pe}{1 - Pe} \quad (10)$$

where $Po = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N w_{ij} \mathcal{C}_{ij}$, which means observed agreement and $Pe = \sum_{i=1}^N \sum_{j=1}^N w_{ij} p_i p_j$ named by expected agreement, N denotes the number of samples. The weighting schemes w_{ij} considered in this article is computed as given, where $w_{ij}=1$ when $i=j$ for $i, j = 1, 2, \dots, R$. M_{ij} denotes to quantitatively represent the degree of difference between two levels.

$$M_{ij} = \max(i, j) - \min(i, j) + 1, w_{ij} = 1 - \frac{M_{ij}}{\max(M_{ij})} \quad (11)$$

The confusion matrix $\mathcal{C}$ and weight coefficients can be found in Fig. 16. Through calculation, the result of $\mathcal{K}$ falls within the range of 0 to 1, at 0.72. Referencing the interpretative guidelines (0.61 to 0.80 indicates substantial agreement), this suggests that the pairwise ranking results predicted by the method described in this paper are in significant consistency with human subjective aesthetic preferences.

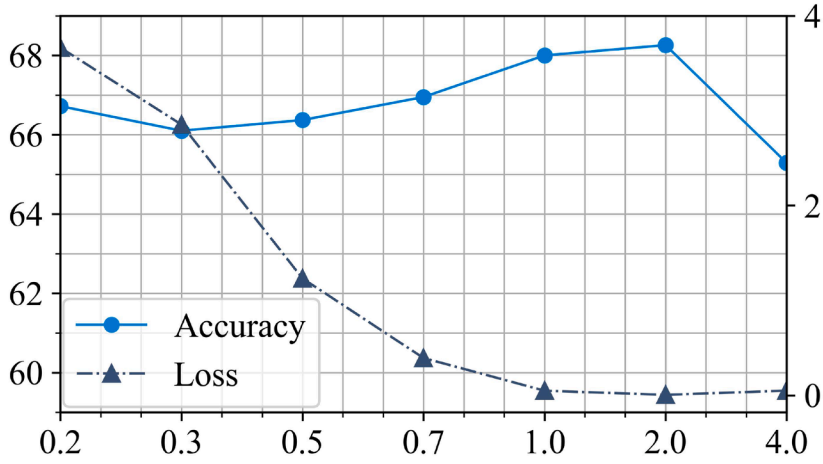
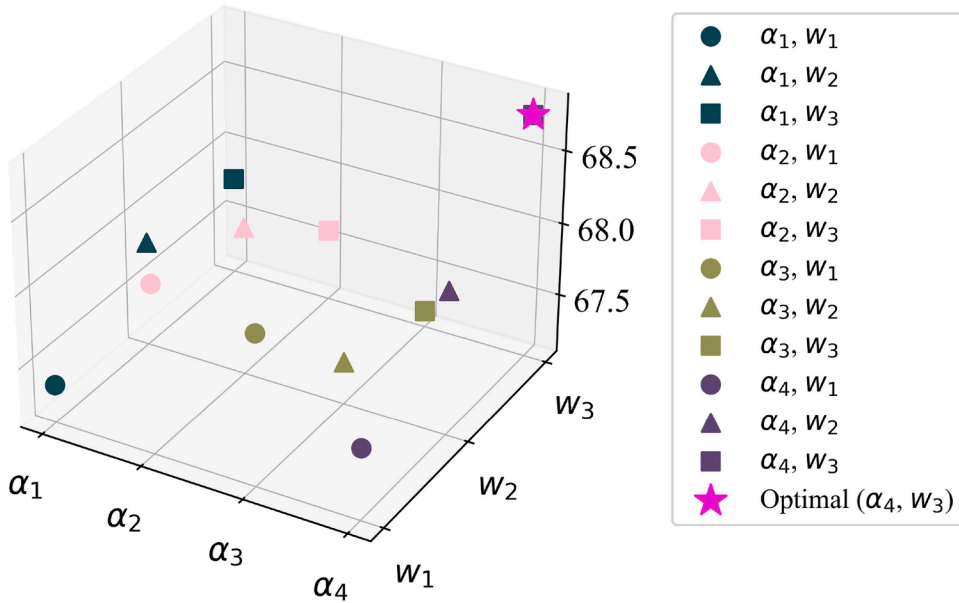
(a) Accuracy - γ - Loss(b) α_j - Accuracy - w_j

Fig. 15. Effects of the hyperparameters on the performance.

5. Implications of our study

5.1. Theoretical implications

It is foreseeable that fashion recommendation systems based on image aesthetic preferences will rapidly become mainstream, as more and more researchers work to narrow the gap between artificial intelligence and human subjective judgment. This shift moves from objective evaluation to a more nuanced understanding of subjective preferences. In this study, we propose a method to compare the fashionability of outfits based on aesthetic assessment, which, to the best of our knowledge, has not been directly explored before. We conducted an initial investigation by developing a custom objective function and constructing a tailored dataset. The theoretical

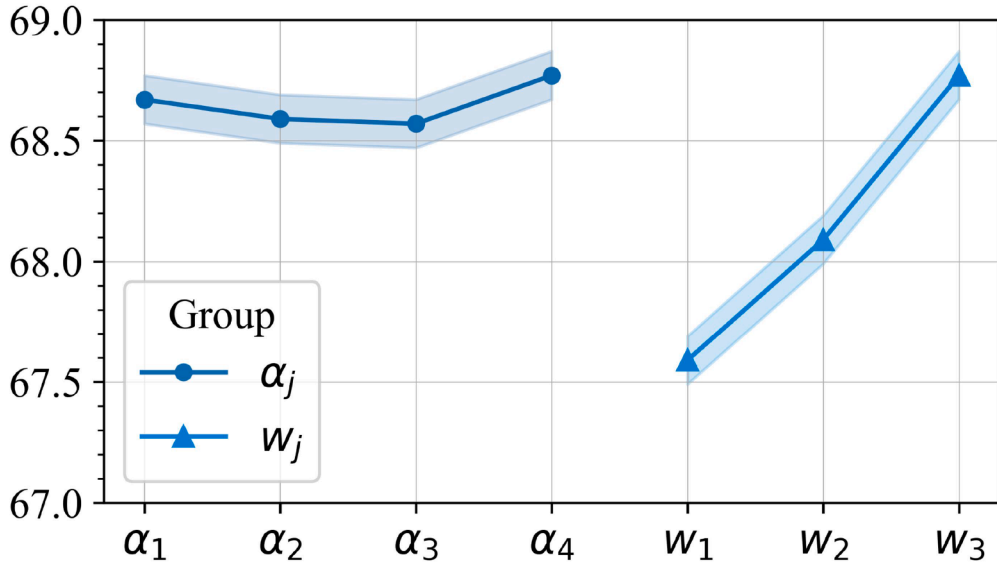
(c) Accuracy range – α_j, ω_j

Fig. 15. (continued).

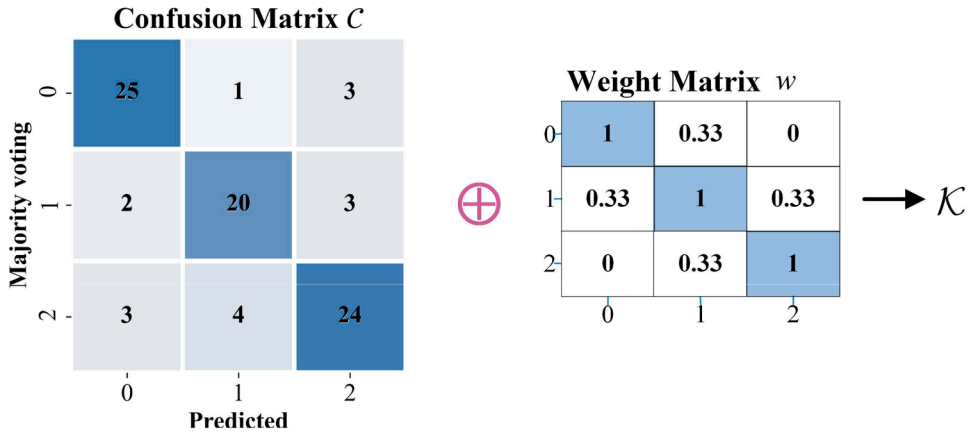


Fig. 16. Confusion matrix and weight coefficients.

significance of our work lies in the following:

- **Relative fashionability:** Our study contributes to the theoretical foundation of fashion ranking by introducing ordinal focal loss combined with relative attribute learning. Unlike traditional methods that rely on absolute labels for evaluating fashionability, our approach shifts the focus to relative comparisons between outfits. This paradigm shift allows for more precise modeling of the subtle, comparative nature of fashion judgments, which are inherently relational rather than absolute. By developing relative attribute learning, we address a gap in the literature where relative rankings of fashionability have been largely underexplored. This perspective aligns more closely with how humans naturally judge fashion, through comparisons rather than isolated assessments.
- **Ordinal focal loss:** Our study presents a novel ordinal focal loss that overcomes the limitations of traditional focal loss in multi-class classification tasks. By incorporating ordinal information, the method enables models to accurately capture the ranking and relationships between categories, improving classification performance, especially in imbalanced data scenarios. This approach broadens the use of focal loss in tasks with ordinal labels, like fashion ranking. Moreover, the softmax-based ordinal loss function can be directly implemented in frameworks like LightGBM, removing the need for transformation techniques such as One-vs-Rest (OVR), making it more efficient and straightforward.

- **Joint local-global feature representation:** Moreover, the incorporation of joint local-global feature representation pushes the boundaries of how visual features are used in fashion assessment. Traditional approaches often fail to capture the complex interactions between clothing items within an outfit by treating each item independently. Our method addresses this limitation by recognizing that the overall aesthetic impression is derived from both item-level (local) and outfit-level (global) interactions. This theoretical advancement contributes to a deeper understanding of fashion aesthetics by modeling not just individual items but their combined effect as an outfit, offering a more holistic and nuanced representation.

5.2. Practical implications

The practical significance of this study's introduction of relative fashionability lies in its alignment with how humans naturally perceive fashion. Specifically, it addresses the following:

- **Outfit recommendations:** The practical significance of studying relative fashionability lies in its ability to provide more accurate outfit recommendations at the final decision-making stage of the user's shopping journey. By focusing on aesthetics, the system can suggest complete outfits, potentially influencing users to purchase entire sets rather than individual items, thus saving time spent on outfit coordination. This approach not only enhances the user experience by offering time-saving solutions but also helps prevent impulsive or misguided purchases caused by poor outfit matching. This contributes to a more informed and satisfying shopping process, reducing unnecessary consumption.
- **Fashionability3k:** The dataset we developed provides the research community with a comprehensive dataset of outfits, paired with relative fashion labels. This dataset is designed specifically to enhance models' abilities to rank fashionability based on subtle differences, making it a valuable resource for both academic and industry applications. As fashion judgments in the real world are often comparative, this dataset allows for more relevant and precise evaluations of model performance in such tasks.

6. Conclusion and future work

In this paper, we show the challenge of quantifying subjective human aesthetic preferences, however, the necessity for fashion recommendation systems to take this into account. Herein, we present an approach of joint categorical and ordinal learning for fashionability grading in outfit images. Conducting ordinal multiclassification tasks with a custom objective function, the proposed joint learning model achieves not only positive classification performance but also shows marked consistency of human aesthetic preferences, leading to an improved fashion analysis that could facilitate aesthetic-based recommendations in e-commerce. Notably, under the comparative relations of relative concepts, previous scholars overlooked the judgment of "similar" pairs, while our method also learns the fashion similarity between two outfits. This is because, in real life, we cannot force users to make an either-or decision. However, the additive value of the proposed method to practical recommendation scenarios still remains to be investigated. Uncovering personal aesthetic preferences requires long-term tracking, and fashion preferences tend to change over time. In Future work, we believe that there is a margin for improvements on our current results. Furthermore, the explainability of the learned aesthetic model is crucial for user trust, especially given the subjective nature of fashion. Techniques to provide intuitive explanations of why certain outfits are recommended or rated higher could significantly increase transparency and value for real-world deployment.

CRediT authorship contribution statement

Ling Ma: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chuhang Zou:** Writing – review & editing, Visualization, Validation, Software, Formal analysis. **Ziyi Guo:** Writing – review & editing, Formal analysis. **Tao Li:** Writing – review & editing, Resources. **Zheng Liu:** Writing – review & editing, Supervision, Funding acquisition. **Fengyuan Zou:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Appendix A. Detailed derivation of gradients and Hessian for custom objective function

The first derivative of the objective function with respect to z_j :

$$\frac{\partial L_{off}}{\partial z_j} = \omega_j y_i \frac{\partial L_{fl}}{\partial p_i} \frac{\partial p_i}{\partial z_j}$$

$$\frac{\partial L_{fl}}{\partial p_i} \frac{\partial p_i}{\partial z_j} = \alpha_j \gamma (1 - p_i)^{\gamma-1} \frac{\partial p_i}{\partial z_j} \log(p_i) - \alpha_i (1 - p_i)^{\gamma} \frac{1}{p_i} \frac{\partial p_i}{\partial z_j}$$

$$\begin{aligned}\frac{\partial L_{fl}}{\partial p_i} \frac{\partial p_i}{\partial z_j} &= \begin{cases} \alpha_i \gamma (1-p_i)^{\gamma-1} \frac{\partial p_i}{\partial z_i} \log(p_i) - \alpha_i (1-p_i)^\gamma \frac{1}{p_i} \frac{\partial p_i}{\partial z_i}, & \text{if } i = j \\ \alpha_j \gamma (1-p_i)^{\gamma-1} \frac{\partial p_i}{\partial z_j} \log(p_i) - \alpha_i (1-p_i)^\gamma \frac{1}{p_i} \frac{\partial p_i}{\partial z_j}, & \text{if } i \neq j \end{cases} \\ \frac{\partial L_{fl}}{\partial p_i} \frac{\partial p_i}{\partial z_j} &= \begin{cases} \alpha_i \left[\gamma (1-p_i)^{\gamma-1} \log(p_i) - (1-p_i)^\gamma \frac{1}{p_i} \right] p_i (1-p_i), & \text{if } i = j \\ \alpha_j \left[\gamma (1-p_i)^{\gamma-1} \log(p_i) - (1-p_i)^\gamma \frac{1}{p_i} \right] (-p_i p_j), & \text{if } i \neq j \end{cases} \\ \frac{\partial L_{fl}}{\partial p_i} \frac{\partial p_i}{\partial z_j} &= \begin{cases} \alpha_i [\gamma (1-p_i)^{\gamma-1} p_i \log(p_i) - (1-p_i)^\gamma] (1-p_i), & \text{if } i = j \\ \alpha_j [\gamma (1-p_i)^{\gamma-1} p_i \log(p_i) - (1-p_i)^\gamma] (-p_j), & \text{if } i \neq j \end{cases} \\ \frac{\partial L_{fl}}{\partial p_i} \frac{\partial p_i}{\partial z_j} &= \begin{cases} \alpha_i [t \gamma p_i - (1-p_i)^{\gamma+1}], & \text{if } i = j \\ -\alpha_i p_j \left[\frac{t \gamma p_i}{1-p_i} - (1-p_i)^\gamma \right], & \text{if } i \neq j \end{cases}\end{aligned}$$

The second derivative of the objective function with respect to z_j :

$$\begin{aligned}\frac{\partial^2 L_{ofl}}{\partial z_j^2} &= \omega_j y_i \left(\frac{\partial}{\partial z_j} \left(\frac{\partial L_{fl}}{\partial p_i} \right) \frac{\partial p_i}{\partial z_j} + \frac{\partial L_{fl}}{\partial p_i} \frac{\partial}{\partial z_j} \left(\frac{\partial p_i}{\partial z_j} \right) \right) \\ \varepsilon &= \frac{\partial}{\partial z_j} \left(\frac{\partial L_{fl}}{\partial p_i} \right) \frac{\partial p_i}{\partial z_j} + \frac{\partial L_{fl}}{\partial p_i} \frac{\partial}{\partial z_j} \left(\frac{\partial p_i}{\partial z_j} \right) \\ \varepsilon &= \begin{cases} \alpha_i \left(-\gamma^2 (1-p_i)^{\gamma-1} p_i \log(p_i) \frac{\partial p_i}{\partial z_i} + \gamma (1-p_i)^\gamma \log(p_i) \frac{\partial p_i}{\partial z_i} + \gamma (1-p_i)^\gamma \frac{\partial p_i}{\partial z_i} + (\gamma+1)(1-p_i)^\gamma \frac{\partial p_i}{\partial z_i} \right), & \text{if } i = j \\ \alpha_j \left(\left[-\gamma(\gamma-1)(1-p_i)^{\gamma-2} p_i \log(p_i) \frac{\partial p_i}{\partial z_j} + \gamma (1-p_i)^{\gamma-1} \log(p_i) \frac{\partial p_i}{\partial z_j} + \gamma (1-p_i)^{\gamma-1} \frac{\partial p_i}{\partial z_j} + \gamma (1-p_i)^{\gamma-1} \frac{\partial p_i}{\partial z_j} \right] (-p_j) \right. \\ \left. - [\gamma (1-p_i)^{\gamma-1} p_i \log(p_i) - (1-p_i)^\gamma] \frac{\partial p_j}{\partial z_j} \right), & \text{if } i \neq j \end{cases} \\ \varepsilon &= \begin{cases} \alpha_i p_i (1-p_i) \left(-\gamma^2 (1-p_i)^{\gamma-1} p_i \log(p_i) + \gamma (1-p_i)^\gamma \log(p_i) + (2\gamma+1)(1-p_i)^\gamma \right), & \text{if } i = j \\ \alpha_j \left(\left[-\gamma(\gamma-1)(1-p_i)^{\gamma-2} p_i \log(p_i) + \gamma (1-p_i)^{\gamma-1} \log(p_i) + 2\gamma (1-p_i)^{\gamma-1} \right] p_i p_j^2 \right. \\ \left. - [\gamma (1-p_i)^{\gamma-1} p_i \log(p_i) - (1-p_i)^\gamma] p_j (1-p_j) \right), & \text{if } i \neq j \end{cases} \\ \varepsilon &= \begin{cases} \alpha_i p_i (1-p_i) \left(-\gamma^2 \frac{t p_i}{1-p_i} + \gamma t + (2\gamma+1)(1-p_i)^\gamma \right), & \text{if } i = j \\ \alpha_j \left(p_i p_j^2 \left[(\gamma-\gamma^2) \frac{t p_i}{(1-p_i)^2} + \frac{t \gamma}{1-p_i} + 2\gamma (1-p_i)^{\gamma-1} \right] - p_j (1-p_j) \left[\frac{\gamma t p_i}{1-p_i} - (1-p_i)^\gamma \right] \right), & \text{if } i \neq j \end{cases}\end{aligned}$$

(Note: let $t = (1-p_i)^\gamma \log(p_i)$)

Appendix B. Part of the RQ3 result

We present partial experimental results from [Section 4.5](#) in [Table A1](#), which primarily include randomly selected image pairs of the test set, predicted outcomes of the model (Predicted), and results from human voting (Majority voting).

Table A1

Part of the human approval test result.

Image pairs (m, n)		Predicted	Majority voting
		p	p
		f	f
		:	f
		p	p
		:	:
		p	p
		p	:
		f	p

Note:

Taking image m as the reference, ‘f’: higher fashionability than n ; ‘p’: lower fashionability than n ; ‘:’: equivalent fashionability.

Data availability

<https://github.com/MLR930/Fashionability3k.git>

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